

Jamie Seoh

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EDUCATION

University of Waterloo

Bachelor of Software Engineering

Sep 2025 – Present

3.95/4.00 (91.2%) GPA

- Top 7.6% in Euclid Math Contest, 2 x Hackathon Winner, Software Engineering Entrance Scholarship

EXPERIENCE

Cerebras Systems

AI Software Engineer Intern | C++, Python, Assembly, Bash, GTest, Pytest

May 2026 – Aug 2026

Toronto, Ontario

- Engineered an end-to-end **hardware fault-injection framework** that simulates crashes, stalls, and slowdowns across the inference pipeline, helping engineers reproduce and diagnose difficult-to-isolate customer failures.
- Reduced false-positive on-call alerts by **84%** by designing a two-tier hardware-fault detection system.
- Built a **KV-cache breakpoint mechanism** that pauses faulted workloads and captures diagnostics for analysis.
- Sped up developer productivity with a static linter for Cerebras assembly language and an interactive dashboard.

SolveXchange

Front-End Developer | HTML, CSS, JavaScript, Flask, SQLAlchemy

Sep 2023 – Jan 2024

Online

- Shipped responsive frontend interfaces serving **100+ active users** on a nonprofit entrepreneurship platform.
- Implemented **JWT**-based authentication and REST APIs with **Flask** for user and forum workflows.

PROJECTS

Quota — DeltaHacks12 First Place | TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed **10,000+**-line codebases in under **3 seconds** using **AST**-based inefficiency detection.
- Cut developer API spend by up to **40%** by providing real-time API cost insights via regex and usage caching.
- Created a React Flow sandbox for budget-aware system planning with **RAG Gemini** chatbot.

Nebula AI | React Native, FastAPI, PostgreSQL, Docker, GCP

- Shipped a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling text extraction and AI-driven summaries.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

Personal CRM | Next.js, FastAPI, PostgreSQL, TailwindCSS

- Developed a full-stack CRM to centralize and prioritize personal & professional relationships at scale.
- Architected backend with **20+ REST** endpoints, enabling pagination & dynamic fetching for growing datasets.
- Improved system reliability through caching layers and rate limiting, reducing server load by **30%**.
- Designed a **scoring** algorithm using relationship recency, frequency, and priority to rank contacts for follow-up.

COMMUNITY

Hack the North

Logistics Organizer

Feb 2026 – Present

Waterloo, Ontario

- Led end-to-end operations for **300+ volunteers** supporting Hack the North, a **1,000+**-participant hackathon.
- Redesigned the volunteer shift-selection system to prevent assignment race conditions and streamline sign-up.
- Oversaw **300+** 3D-printed keychains from design to production given time and material constraints.

TECHNICAL SKILLS

Languages: Python, C/C++, Bash, TypeScript & JavaScript, HTML & CSS, SQL, Java, R

Frameworks and Libraries: PyTorch, NumPy, React.js, React Native, Next.js, Svelte, FastAPI, Flask, Tailwind CSS

Technologies: Docker, Git/GitHub, GCP, Azure, AWS, MongoDB, PostgreSQL, Linux, Figma, Fusion 360